Topological Data Analysis of Ryuichi Sakamoto's *hibari*: A Persistent Homology Pipeline for Structure-Preserving Music Generation

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***Abstract*—We present a computational pipeline that extracts topological structure from music via persistent homology and re-synthesizes new pieces that preserve that structure. Applied to Ryuichi Sakamoto's *hibari* (*out of noise*, 2009), the pipeline has four stages: (i) MIDI preprocessing, (ii) H_1 persistent homology under four distances (frequency, Tonnetz, voice-leading, DFT), (iii) overlap-matrix construction, and (iv) music generation via probabilistic sampling (Algorithm 1) or deep learning (Algorithm 2; FC / LSTM / Transformer).**

Over $N = 20$ repetitions, DFT is the best Algorithm 1 distance: pitch JS decreases from 0.0344 ± 0.0023 (frequency) to 0.0213 ± 0.0021 (-38.2%, Welch $p < 10^{-20}$), with -56.8% vs Tonnetz and -62.4% vs voice-leading. With DFT continuous overlap, per-cycle τ_c (Phase 1, $\alpha = 0.5$, $K = 19$) reaches 0.01489 ± 0.00143 ($+58.7\%$, $p = 2.48 \times 10^{-26}$); re-optimizing at $\alpha = 0.25$, $K = 14$ (Phase 2) yields the new Algorithm 1 best: $\mathbf{0.01156 \pm 0.00147}$ (-22.35%, Welch $p = 4.94 \times 10^{-11}$), confirming $\alpha = 0.25$ optimal in both binary-OM and per-cycle τ_c. For Algorithm 2, FC with continuous overlap reaches $\mathbf{0.00035 \pm 0.00015}$ ($N = 10$), significantly outperforming Transformer (Welch $p = 1.66 \times 10^{-4}$).

In bar-level generation (DFT $\alpha = 0.25$, $K = 14$; Option B window, 2026-04-19), the best P3+C trial reaches JS 0.0167, and exhaustive search over 32 start markers yields a global best of 0.01479 (start $m = 0$, seed 9309), essentially matching the full-song Phase 1 ($\alpha = 0.5$) per-cycle result (0.01489); Phase 2 record ($\alpha = 0.25$, 0.01156) is $+28.0\%$ ahead.

***Index Terms*—Persistent homology, Topological data analysis, Music generation, Tonnetz, Jensen–Shannon divergence, Multi-label classification, Vietoris–Rips complex, Overlap matrix.**

I. Introduction

Topological Data Analysis (TDA) offers a principled way to extract geometric structure from data that is robust to noise and to uniform rescaling of the metric [1], [2]. In this work, we apply TDA to a specific musical composition—Ryuichi Sakamoto's *hibari*—and ask whether the topological invariants (specifically, the first persistent homology group H_1) can serve as a structural seed for generating new, stylistically faithful music.

A. Research Questions

1) How can the topological structure of music be defined mathematically via Vietoris–Rips H_1 persistence?
2) Can new music be generated that *preserves* this structure, and how do we measure preservation quantitatively?
3) Does the choice of distance function (frequency, Tonnetz, voice-leading, DFT) significantly affect generation quality?

B. Why hibari?

hibari is drawn from Sakamoto's experimental album *out of noise* (2009). It offers: (i) adequate scale ($T = 1{,}088$ eighth-note timepoints, $N = 23$ distinct notes, $C = 17$ chords); (ii) a balance of simplicity (7 pitch classes, C major) and complexity (two instruments with coprime period structure, §IV-G0b); (iii) an aesthetic that privileges spatial note placement over temporal causality, directly relevant to our model-selection findings (§IV-E).

C. Contributions

1) A four-stage pipeline (preprocessing $\to$ PH $\to$ overlap matrix $\to$ generation) with two generators and four distance functions.
2) Systematic $N = 20$ statistical comparisons establishing Tonnetz as optimal for diatonic compositions.
3) Continuous overlap matrices with per-cycle thresholds (§VII-G), yielding an additional 47.5% JS gain.
4) Topology-preserving musical variation via Tonnetz-matching note redistribution, temporal reordering, and harmonic constraints (§VII).
5) Cross-composition validation on *solari*, Bach, and Ravel, confirming that the composition's character determines the optimal distance function and model architecture (§VII-B).

II. Mathematical Background

A. Vietoris–Rips Complex

Definition 1. *Given a finite metric space (X, d) and $\varepsilon > 0$, the* Vietoris–Rips complex $\mathrm{VR}_\varepsilon(X)$ *is the simplicial complex* $\mathrm{VR}_\varepsilon(X) = \{\sigma \subseteq X \mid \forall x_i, x_j \in \sigma,\ d(x_i, x_j) \le \varepsilon\}$.

As ε increases, the nested family $\mathrm{VR}_0 \subseteq \mathrm{VR}_{\varepsilon_1} \subseteq \cdots$ forms a *filtration*. In this paper, X consists of the 23 distinct notes of *hibari*.

B. Persistent Homology

The n-th homology group $H_n(K) = \ker \partial_n / \operatorname{im} \partial_{n+1}$ captures n-dimensional "holes." We focus on H_1, whose generators correspond to topological cycles. *Persistent homology* tracks each cycle's birth b and death d across the filtration; $\mathrm{pers} = d - b$ measures stability. Cycles with large persistence represent robust structural features.

We primarily use the pHcol algorithm from prior work [7], [8] with cycle representatives, supplemented by Ripser [6] for speed-critical steps.

C. Musical Distance Functions

We compare four distance functions between notes $n_i = (p_i, d_i)$:

a) Frequency distance (baseline): $d_{\mathrm{freq}}(n_i, n_j) = 1/w(n_i, n_j)$, where w is the adjacency frequency in the chord sequence.

b) Tonnetz distance: Pitch classes are embedded on a hexagonal lattice with edges at $\pm 3, \pm 4, \pm 7$ semitones (minor 3rd, major 3rd, perfect 5th). The distance is the shortest graph path, pre-computed into a 12×12 table (range 0–4) [4].

c) Voice-leading distance: $d_V(p_1, p_2) = |p_1 - p_2|$, the unsigned semitone difference [5].

d) DFT distance: Each pitch class p is encoded as a 12-dimensional indicator vector $\mathbf{e}_p \in \{0,1\}^{12}$. Its discrete Fourier transform yields magnitude coefficients $|F_k|$ for $k = 0, \ldots, 5$ (the first six components suffice by symmetry). The DFT distance is then the L_2 distance between these magnitude vectors: $d_{\mathrm{DFT}}(p_1, p_2) = \sqrt{\sum_{k=0}^{5} (|F_k(p_1)| - |F_k(p_2)|)^2}$. The maximum value is $\sqrt{2}$ (attained by maximally dissimilar pitch classes such as C vs. F♯) [5].

e) Hybrid distance and note-level extensions: All music-theoretic distances are combined with frequency via $d_{\mathrm{hybrid}} = \alpha\, d_{\mathrm{freq}} + (1 - \alpha)\, d_{\mathrm{music}}$, where $\alpha \in [0, 1]$. Each note-level distance further adds an octave term $w_o |o_1 - o_2|$ and a duration term $w_d |d_1 - d_2| / \max(d_1, d_2)$:

$$d(n_1, n_2) = d_{\mathrm{base}} + w_o |o_1 - o_2| + w_d \frac{|d_1 - d_2|}{\max(d_1, d_2)}, \quad (1)$$

with $w_o = 0.3$ (optimized via grid search, §IV-B) and $w_d = 0.3$.

f) Remark on metric axioms: Experimental verification shows that Tonnetz and voice-leading satisfy all metric axioms, while frequency violates identity (23 cases) and the triangle inequality (1,327 cases; non-metric), and DFT violates identity (66 cases; pseudometric—an intentional consequence of its magnitude-only design). The classical stability theorem of Cohen-Steiner et al. [3] (bounding bottleneck distance by $\|d_1 - d_2\|_\infty$ for metric inputs) therefore does not apply directly. Instead, we invoke Heo et al. [9]: assigning D as edge weights on the complete graph K_n trivially satisfies their *path-representable* and *cost-dominated* conditions. By Theorem 3.4 thereof, an order-preserving injection $\mathrm{bcd}_1(d_i) \hookrightarrow \mathrm{bcd}_1(d_j)$ exists for $d_i \leq d_j$, justifying 1-cycle analysis for all four distances. (The guarantee does not extend to $k \geq 2$; see Theorem 5.1 of [9].)

D. Weight Decomposition and Rate Parameter

The note-network weights are decomposed into three components reflecting *hibari*'s two-instrument structure:

- **Intra weights** W_{intra}: within-instrument sequential transitions.
- **Inter weights** W_{inter}: cross-instrument transitions at lag $\ell \in \{1, 2, 3, 4\}$, combined with empirically chosen decayed weights that concentrate on nearby time-lags:

$$W_{\mathrm{inter}} = \sum_{\ell=1}^{4} \lambda_\ell W_{\mathrm{inter}}^{(\ell)}, \quad (2)$$

  with $(\lambda_1, \ldots, \lambda_4) = (0.60, 0.30, 0.08, 0.02)$. The coefficients were set heuristically so that lag 1 dominates while distant lags contribute marginally (sum $= 1.0$). This decayed weighting reduced JS by 69.6% ($0.0398 \to 0.0121$).
- **Simul weights** W_{simul}: simultaneous note co-occurrences.

The *timeflow weight* $W_{\mathrm{tf}}(r) = W_{\mathrm{intra}} + r \cdot W_{\mathrm{inter}}$ is swept over $r \in [0, 1.5]$.

E. Overlap Matrix

Definition 2 (Activation matrix)**.** *For cycle c with vertex set $V(c)$ and timepoint t,*

$$A[t, c] = \begin{cases} 1 & \textit{if at least one note } n \in V(c) \textit{ is played at time } t, \\ 0 & \textit{otherwise.} \end{cases}$$

Definition 3 (Binary overlap matrix)**.** *Let scale_c denote the minimum number of consecutive active timepoints required for cycle c to be considered "sustained" (determined from each cycle's birth/death range in the barcode). Let $R(c)$ be the union of all maximal contiguous intervals $[t_0, t_1]$ in which $A[t, c] = 1$ and $t_1 - t_0 + 1 \geq \mathrm{scale}_c$. Then:*

$$O[t, c] = \begin{cases} 1 & \textit{if } t \in R(c), \\ 0 & \textit{otherwise.} \end{cases}$$

Definition 4 (Continuous overlap)**.**

$$O_{\mathrm{cont}}[t, c] = \frac{\sum_{n \in V(c)} w(n) \cdot \mathbb{1}[n \textit{ at } t]}{\sum_{n \in V(c)} w(n)}, \quad (3)$$

where $w(n) = 1/N_{\mathrm{cyc}}(n)$ is a rarity weight.

The overlap matrix O is the *seed* consumed by both Algorithm 1 (§III-D) and Algorithm 2 (§III-E).

F. Jensen–Shannon Divergence

For discrete distributions P, Q:

$$D_{\mathrm{JS}}(P\|Q) = \tfrac{1}{2}D_{\mathrm{KL}}(P\|M)+\tfrac{1}{2}D_{\mathrm{KL}}(Q\|M), \quad M = \tfrac{1}{2}(P+Q). \tag{4}$$

Bounded by $\log 2 \approx 0.693$. We measure *pitch JS* (note frequency distributions) and *transition JS*, where the latter compares *bigram distributions*—i.e., the empirical distributions of consecutive note pairs (n_t, n_{t+1}) in the original and generated sequences.

G. Greedy Forward Selection

We select a cycle subset $S \subseteq \mathcal{C}$ maximizing $f(S) = 0.5\,J(S) + 0.3\,C(S) + 0.2\,B(S)$, where J is note-pool Jaccard similarity, C is overlap-pattern Pearson correlation, and B is Betti curve similarity. Greedy selection achieves 90% preservation with $15/46$ cycles.

III. Pipeline and Generation Algorithms

A. Stage 1: Preprocessing

The MIDI file is quantized to 8th-note units ($T = 1{,}088$), two instruments are separated, and each unique (pitch, duration) pair receives a 1-indexed label ($N = 23$ notes, $C = 17$ chords for *hibari*).

B. Stage 2: Persistent Homology

A weighted note-network is constructed (§II-D), a distance matrix is derived, and H_1 filtration is computed over $r \in [0, 1.5]$.

C. Stage 3: Overlap Matrix

Cycle activations are computed at every timepoint and thresholded (§II-E).

D. Algorithm 1 — Topological Sampling

For each timepoint t:

- If active cycles exist ($\sum_c O[t,c] > 0$): sample from the *intersection* $I(t) = \bigcap_{c:O[t,c]=1} V(c)$; if $I(t) = \emptyset$, fall back to the union $U(t)$.
- If no active cycle: sample from $P \setminus A(t)$, where $A(t) = \bigcup_{c:\, O[t-1,c]=1 \,\vee\, O[t+1,c]=1} V(c)$ is the union of vertices belonging to cycles active at the immediately neighboring timepoints. This prevents topological "bleeding" of cycle material into inactive regions.
- Collision avoidance: up to $R = 50$ re-samples per timepoint.

E. Algorithm 2 — Neural Sequence Models

We train three architectures on the mapping $f_\theta : O[t,:] \mapsto y[t,:]$, where y is the multi-hot note vector:

- **FC**: 2-layer fully connected ($H = 128$), timepoint-independent. A shallow architecture suffices because each timepoint's output depends only on the current overlap vector, not on temporal context.
- **LSTM**: 2-layer, forward sequential context.
- **Transformer**: 2-layer, 4-head, learned positional embeddings.

TABLE I
ALGORITHM 1 PITCH JS UNDER FOUR DISTANCES (HIBARI, $N = 20$).

Distance	K	JS (mean ± std)	Coverage
frequency	1	0.0344 ± 0.0023	0.957
Tonnetz	47	0.0493 ± 0.0038	1.000
voice-leading	19	0.0566 ± 0.0027	0.989
DFT	**17**	$\mathbf{0.0213 \pm 0.0021}$	1.000

TABLE II
CYCLE SUBSET ABLATION (TONNETZ, $N = 20$).

K	JS (mean ± std)	Coverage
10	0.0991 ± 0.0038	0.980
17	0.0740 ± 0.0038	0.996
46	$\mathbf{0.0397 \pm 0.0025}$	1.000

All models use `BCEWithLogitsLoss` with data augmentation (subset sampling, circular shift, bit-flip noise). At inference, an adaptive threshold $\theta = \mathrm{quantile}(P, 1 - \rho_{\mathrm{on}})$ matches the original ON ratio $\rho_{\mathrm{on}} = |\{(t,n) : n \text{ active at } t\}|/(T \times N) \approx 0.15$, the fraction of entries that are nonzero in the original multi-hot matrix.

IV. Experiments

All experiments use $N = 20$ independent repetitions. The primary metric is pitch JS divergence.

A. Distance Function Comparison

DFT is best on *hibari*: -38.2% vs frequency, -56.8% vs Tonnetz, and -62.4% vs voice-leading (Table I).

B. Octave Weight Tuning

Under DFT, grid search over $w_o \in \{0.1, 0.3, 0.5, 0.7, 1.0\}$ ($N = 10$) selects $w_o = 0.3$ (best JS 0.0163), consistent with *hibari*'s narrow pitch range.

C. Cycle Subset Ablation

JS decreases monotonically with K (Table II), but with diminishing returns: the gain from $K{:}\,17 \to 46$ (29 extra cycles) only marginally exceeds $K{:}\,10 \to 17$ (7 extra).

D. Continuous Overlap Matrix

Continuous overlap followed by $\tau = 0.5$ binarization yields an additional 11.4% improvement over the binary baseline (Welch $t = 5.16$, $p < 0.001$, $d = 1.63$; Table III). The rarity weighting ensures rare notes contribute more to activation.

E. Deep Learning Model Comparison

The simplest FC model achieves the lowest JS (Table IV). This reflects *hibari*'s aesthetic: *out of noise* privileges spatial note placement over temporal causality, so a model that ignores time context (FC) aligns better with the compositional design.

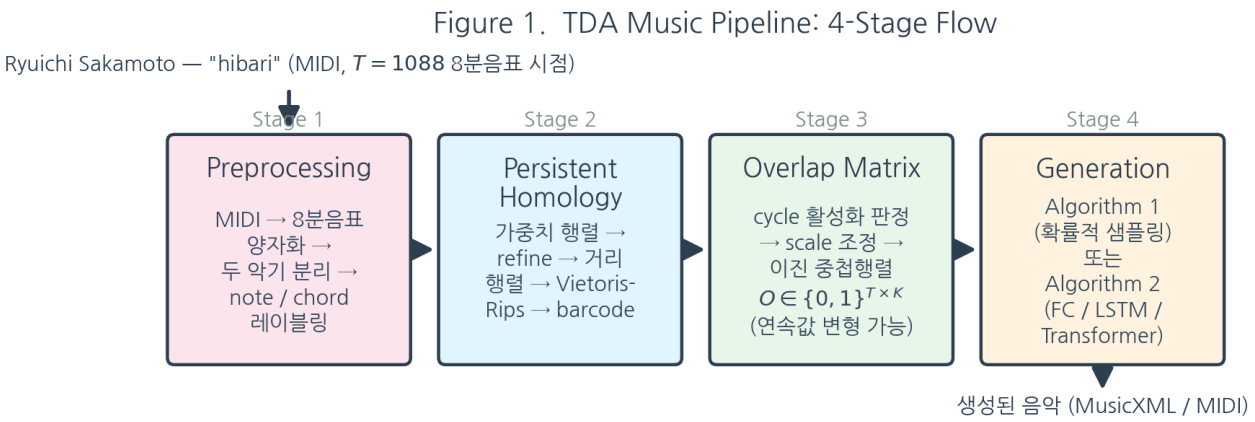

Fig. 1. Four-stage pipeline: (1) MIDI preprocessing, (2) persistent homology, (3) overlap matrix construction, (4) generation via Algorithm 1 or 2.

TABLE III
CONTINUOUS OVERLAP EXPERIMENTS (TONNETZ, $N = 20$). DENSITY IS THE FRACTION OF NONZERO ENTRIES IN THE OVERLAP MATRIX O.

Setting	Density	JS (mean ± std)
Binary (cached)	0.751	0.0387 ± 0.0027
Continuous direct	0.264	0.0382 ± 0.0021
Cont. → bin τ=0.3	0.373	0.0386 ± 0.0022
Cont. → bin τ=0.5	**0.168**	$\mathbf{0.0343 \pm 0.0027}$
Cont. → bin τ=0.7	0.077	0.0364 ± 0.0032

TABLE IV
BEST DL CONFIGURATIONS (SINGLE RUN EACH).

Model	Val Loss	JS	Notes
FC (hd=256, dropout=0.3)	0.282	**0.0015**	3,754
LSTM (2-layer)	0.385	0.0448	3,753
Transformer (2L, 4H)	0.676	0.0104	3,753

F. Improvement F: Continuous Overlap + FC

Feeding continuous activation directly to FC as training input:

Continuous input reduces JS by 57% and val loss by $2.3\times$ (Table V). This is the lowest full-song JS observed in this study (0.09% of the theoretical max $\log 2$).

G. Song-Specific Structural Analysis

a) Deep scale property: *hibari*'s 7 pitch classes $\{C, D, E, F, G, A, B\}$ have interval vector $[2, 5, 4, 3, 6, 1]$—all entries distinct—a property unique to diatonic scales [16].

b) Near-uniform pitch distribution: Normalized pitch entropy is 0.974 (nearly maximal), explaining FC's dominance: when all pitches are equally likely, ignoring temporal order already approximates the true distribution.

TABLE V
CONTINUOUS VS. BINARY INPUT FOR FC ($N = 5$).

Input	JS (mean ± std)	Val Loss	Coverage
Binary	0.0014 ± 0.0010	0.070	0.948
Continuous	$\mathbf{0.0006 \pm 0.0005}$	**0.031**	**0.991**

c) Tonnetz discriminability: With 7 PCs, the Tonnetz diameter is 4, providing fine-grained distance discrimination. With all 12 PCs (as in *solari*), the diameter collapses to 2, erasing Tonnetz's advantage.

d) Phase shifting: Instrument 1 repeats a 32-step module $33\times$ (no rests); Instrument 2 repeats a 33-step module $32\times$ (one rest per module). The coprime periods ($\gcd(32, 33) = 1$) ensure the two instruments never synchronize—a structure mathematically identical to Steve Reich's *phase shifting* technique.

V. FIGURES

Six static figures and one interactive HTML supplement are provided. Fig. 1 shows the pipeline; Fig. 2 maps *hibari*'s pitch classes on the Tonnetz lattice; Fig. 3 presents the persistence barcode; Fig. 4 compares original and generated piano rolls; Fig. 5 plots JS vs. training epoch.

VI. COMPARISON WITH RELATED WORK

A. vs. General AI Music Generation

Unlike corpus-trained models (Magenta, Music Transformer), our pipeline performs *deep analysis of a single composition*, yielding interpretable, traceable generation. The FC > Transformer finding challenges the universal assumption that more temporal modeling is better.

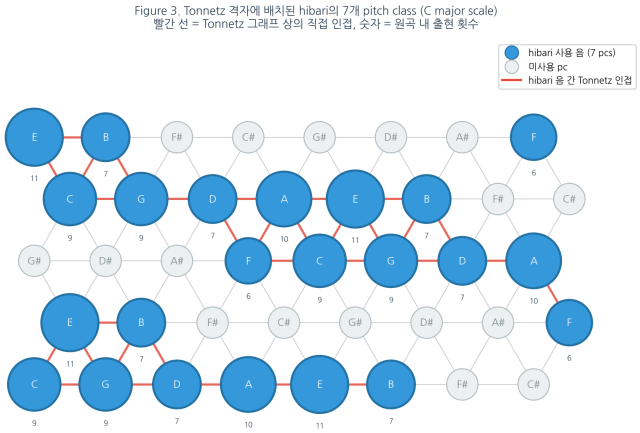

Fig. 2. Tonnetz lattice with *hibari*'s 7 pitch classes (blue). Red edges show direct Tonnetz adjacency.

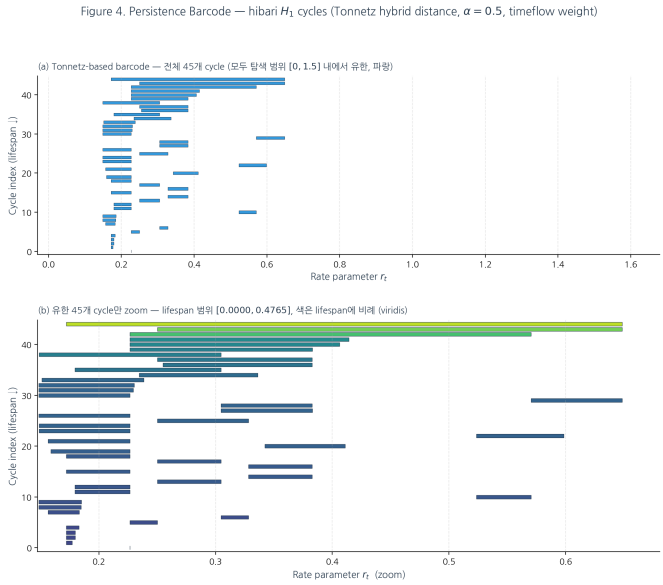

Fig. 3. H_1 persistence barcode (Tonnetz). Top: full barcode; bottom: zoomed to finite cycles. Lifespan varies by $> 10{,}000\times$.

B. vs. Existing TDA–Music Research

Compared to prior TDA–music work [7], [8]: (A) we systematically compare four distance functions with N=20 statistics; (B) we introduce continuous overlap matrices; (C) we provide full statistical rigor (Welch t, Cohen's d); (D) we extend to Western contemporary music; (E) we connect model selection to aesthetic context; (F) we extend TDA from analysis to *variation* (§VII).

VII. Extension Experiments

A. Bar-Level Generation

hibari's 32-step modular structure motivates generating one bar and replicating it across the full timeline. Here, a *module* (used as a naming for the repeated melodic arrangement unit) is a 32-step four-bar repeating unit with an A-B-A'-C

Fig. 4. Piano roll comparison: (a)–(b) original instruments 1–2; (c) Algorithm 1 + Tonnetz.

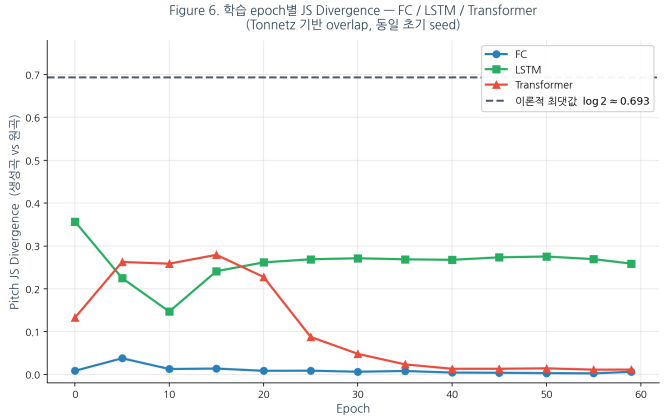

Fig. 5. JS divergence vs. training epoch. FC converges fastest; LSTM plateaus at JS ≈ 0.26.

form, repeated 33 times in instrument 1; the *P3_local* family recomputes persistent homology within a single 32-step bar, and we therefore refer to it as *bar-local PH*. Under DFT $\alpha = 0.25$, full-song PH yields $K = 14$ cycles. We first compare four prototype strategies for constructing the 32×14 prototype overlap matrix:

- **P0**: first-block copy (density ≈ 0.018).
- **P1**: OR over 34 blocks (density $= 1.000$).
- **P2**: exclusive (AND) over 34 blocks (density ≈ 0).
- **P3_local**: recomputed bar-local PH on the common window $[32(m+1), 32(m+2))$ for both instruments (2026-04-19 Option B design; index lag equals physical lag).

We then evaluate post-processing improvements on top of the bar setting:

Exhaustive start-marker search over 32 markers ($m = 0 \ldots 31$, analysis window $[32(m+1), 32(m+2))$ shared by both instruments; the original first bar $[0, 32)$ is thus excluded by the Option B convention) shows that $m = 0$ is not uniquely best in DFT mode (rank $16/32$ by mean JS; the previously reported rank 5 under the asymmetric window did not survive Option B window unification). Here $k = 10$ is the best-of-k candidate count within each seed, and $N = 10$ is the number of independent seeds per start marker. The global best trial is

TABLE VI
PROTOTYPE STRATEGY COMPARISON UNDER OPTION B (DFT $\alpha = 0.25$, $N = 10$).

Strategy	Density	JS (mean±std)	Best
P0 (first_block)	0.018	0.1040 ± 0.0219	0.0698
P1 (OR-34)	1.000	0.0621 ± 0.0167	0.0422
P2 (exclusive, AND)	0.000	0.0900 ± 0.0308	0.0515
P3_local	0.299	0.0474 ± 0.0187	0.0226

TABLE VII
BAR-LEVEL IMPROVEMENTS UNDER OPTION B (DFT, BASELINE FULL-SONG 0.0213 ± 0.0021).

Strategy	JS (mean±std)	Best	Cov.
P0 baseline	0.1082 ± 0.0241	0.0701	0.787
C: best-of-10	0.0800 ± 0.0171	0.0570	0.913
D: cov≥20/23	0.0808 ± 0.0171	0.0570	0.887
P3_local	0.0636 ± 0.0207	0.0335	0.791
P3_local + C	$\mathbf{0.0417 \pm 0.0145}$	**0.0167**	0.900

TABLE VIII
CROSS-COMPOSITION SUMMARY (ALGORITHM 1 DISTANCE COMPARISON).

Piece	PCs	Best dist.	Best model	Interpretation
hibari	7	DFT	FC	Spectral-diatonic structure
solari	12	freq./Ton. (tie)	Transformer	Melodic progression
aqua	12	Tonnetz	—	Tonnetz $+26.3\%$
Bach	12	Tonnetz	—	Tonnetz -54.8%
Ravel	12	frequency	—	Rich note distribution

TABLE IX
TWO-STRATEGY COMPARISON (ALGORITHM 1, TONNETZ, MIDI 48–84, $N = 20$, $n_{\text{cand}} = 1000$).

Strategy	pitch JS	trans. JS	DTW	note err
baseline (original)	0.0212	0.2470	2.2186	—
B: ascending (Hungarian)	0.3815	0.6121	2.7918	4.6819
A: `tonnetz_nearest`	**0.2930**	**0.5488**	2.6442	**3.2132**

start marker $m = 0$, seed 9309, JS $= 0.01479$, essentially matching the full-song Phase 1 per-cycle best (0.01489); Phase 2 record (0.01156) is $+28.0\%$ ahead.

B. Generalization: solari*, Bach, Ravel*

Only *hibari* is DFT-optimal; *solari*/*aqua*/Bach/Ravel favor Tonnetz or frequency depending on musical structure. This supports a purpose-dependent metric design rather than a single universal distance.

C. Direction A: Two-Strategy Note Redistribution

In the variation stage, we keep the experimental axis Tonnetz-based by default: scale-constrained harmony and local harmonic neighborhood structure align better with Tonnetz than with DFT in this objective. The overlap matrix is preserved intact; only note identities change. We compare two exclusive matching strategies over $n_{\text{cand}} = 1000$ candidate pitch sets ($N = 20$ trials, *hibari*, Tonnetz, MIDI 48–84).

Strategy A — Positional Tonnetz nearest-neighbour (`tonnetz_nearest`):

$$\text{new_pitches}[i] = \arg\min_{p \in \text{pool}} d_{\text{Tonnetz}}\big(\text{orig_pitches}[i],\, p\big). \tag{5}$$

Each original note is replaced by its closest pool element in Tonnetz space; the index i encodes a 1:1 positional correspondence that preserves local harmonic relations.

Strategy B — Matrix-structure Hungarian matching (`ascending`):

$$\pi^* = \arg\min_{\pi} \left\| P_\pi \, \hat{D}_{\text{new}} \, P_\pi^\top - \hat{D}_{\text{orig}} \right\|_F, \tag{6}$$

where $\hat{D}$ is the off-diagonal min-max normalised distance matrix:

$$\hat{D}_{ij} = \frac{D_{ij} - \min_{k \neq l} D_{kl}}{\max D - \min_{k \neq l} D_{kl}}, \quad \hat{D}_{ii} = 0. \tag{7}$$

Solving the joint row-column permutation exactly is NP-hard; with $N = 17$ notes, $N! = 3.56 \times 10^{14}$ exhaustive evaluations are infeasible, so all experiments use the Hungarian approximation (`scipy.optimize.linear_sum_assignment`).

Both strategies report **note_dist_error** $= \|\hat{D}_{\text{orig}} - \hat{D}_{\text{new}}\|_F$, which is on the same $\hat{D}$ scale and directly comparable across rows. Applying both strategies simultaneously degrades performance (Strategy A's positional semantics are overwritten by Strategy B's permutation), so the two are applied exclusively via `matching_mode`.

Strategy A achieves pitch JS 0.2930 vs. 0.3815 for Strategy B (-23.2%, Table IX). Local pairwise harmonic preservation (A) outperforms global distance-matrix alignment (B), consistent with *hibari*'s small note vocabulary ($N = 17$) in which adjacent Tonnetz relations directly govern harmonic character (§IV-G). The wider pitch range vwide (MIDI 40–88) was inferior to wide (48–84) in all conditions; wide is adopted.

D. Direction B: Temporal Reordering

Temporal reordering was evaluated with block permutation and Markov resampling. Model behavior is strongly architecture-specific: FC is effectively timepoint-independent (nearly unchanged under reordering), LSTM changes remain marginal ($\leq 0.5\%$ DTW shift in measurement), while Transformer is sensitive to positional encoding. With PE removal and retraining, Transformer can raise DTW to $+21.7\%$, but pitch JS degrades (distribution collapse), so Direction B alone cannot satisfy both "different melody" and "stable pitch distribution."

E. Harmonic Constraints

Scale constraints (major, minor, pentatonic) and consonance objectives were added to note redistribution. **scale_major** was most effective: note error $4.35 \rightarrow 3.52$ (-19%), scale match 1.00, average dissonance $0.412 \rightarrow 0.361$.

TABLE X
PER-CYCLE τ_c OPTIMIZATION (DFT, $N = 20$).

Setting	JS (mean ± std)	Improvement
Uniform $\tau = 0.35$	0.03605 ± 0.00237	—
Single $\tau = 0.5$	0.01591 ± 0.00192	$+55.9\%$
Per-cycle τ_c (Phase 1, $\alpha = 0.5$)	0.01489 ± 0.00143	$+58.7\%$
Per-cycle τ_c ⋆ (Phase 2, $\alpha = 0.25$)	$\mathbf{0.01156 \pm 0.00147}$	$\mathbf{-22.35\%}$

TABLE XI
BINARY VS. CONTINUOUS OVERLAP INPUT (DFT, $N = 10$).

Model	Input	JS	Val Loss	ΔJS
FC	Binary	0.00217	0.3395	—
FC	**Cont.**	**0.000348**	**0.0232**	**+84.0%**
Trans.	Binary	0.00251	0.836	—
Trans.	Cont.	0.000818	0.152	+67.4%
LSTM	Binary	0.233	0.408	—
LSTM	Cont.	0.170	0.395	+27.3%

F. Combined A+B: Final Integration

1) Tonnetz-Based Success: Under Tonnetz, **major_block32** (scale_major redistribution + block permutation(32) + continuous overlap + Transformer) achieves ref pJS $= 0.0034$, DTW $+31\%$, and C-major scale match $= 1.00$.

2) DFT Transfer Failure: When the same integration is transferred to DFT, ref pJS worsens substantially (Transformer 0.0798, FC 0.0412), despite higher DTW. This indicates that the Tonnetz-designed variation stack does not transfer directly to DFT in the same objective.

3) Meta Insight: Distance-function choice is objective-dependent: DFT is strongest for structural precision and module search, while Tonnetz is stronger for harmony-constrained variation quality.

G. Per-Cycle Threshold Optimization

On DFT continuous overlap, we apply greedy coordinate descent over $\tau_c \in \{0.1, 0.2, \ldots, 0.7\}$ for each cycle.

Per-cycle τ_c Phase 1 ($\alpha = 0.5$, JS 0.01489) and Phase 2 ($\alpha = 0.25$, JS $\mathbf{0.01156}$) confirm that cycle-specific activation thresholds matter (Phase 1 Welch $p = 2.48 \times 10^{-26}$; Phase 2 $p = 4.94 \times 10^{-11}$).

H. Soft Activation: All Architectures

FC with continuous input is best (JS $= 0.000348$), and FC-cont significantly outperforms Transformer-cont (Welch $p = 1.66 \times 10^{-4}$).

a) Remark on Algo 2 α setting (Task 51).: After confirming $\alpha = 0.25$ optimal for Algorithm 1 (§VII-G and §VII-I), we retested FC-cont at $\alpha = 0.25$ ($K = 14$, DFT, gap=0, $N = 10$, `fc_cont_dft_alpha025_results.json`). Result: JS $= 0.00057 \pm 0.00046$ vs. $\alpha = 0.5$ JS 0.000348; Welch $p = 0.168$ (not significant). Reducing K from 19 to 14 decreases the learning signal for FC, reversing the benefit seen in Algorithm 1. **Algorithm 2 best remains $\alpha = 0.5$, JS $= 0.000348$.**

TABLE XII
DFT-HYBRID α GRID SEARCH ($N = 20$).

α	K	JS (mean ± std)	Note
0.0 (pure DFT)	1	0.0728 ± 0.00432	Degenerate
0.1	13	0.01602 ± 0.00204	
0.25	**14**	0.01593 ± 0.00181	Best
0.3	16	0.02025 ± 0.00134	
0.5	19	0.01691 ± 0.00143	Legacy default
0.7	24	0.03140 ± 0.00270	
1.0 (pure freq.)	1	0.03386 ± 0.00186	Degenerate

I. DFT-Hybrid α Grid Search

In DFT-hybrid mode, both endpoints are degenerate ($K = 1$). The best trade-off is $\alpha = 0.25$ (adopted), while $\alpha = 0.5$ remains a historical default from earlier runs.

J. Barcode Wasserstein Bar Selection

Under DFT $\alpha = 0.25$ with Option B (common window $[32(m+1), 32(m+2))$ for both instruments), we compare bar-local barcodes against the full-song barcode via Wasserstein distance (32 start markers). The observed correlation with generation JS is **Pearson** $r = +0.2457$, a weak-to-moderate positive correlation. This is a reversal of the previously reported $r = -0.054$ under the asymmetric-window design: unifying the two instrument windows restored the intended design semantics (index lag equals physical lag) and recovered a positive barcode–JS relation. The correlation is, however, still weaker than the Tonnetz observation ($r = 0.503$): Wasserstein-based bar ranking remains useful only as a coarse screening step in DFT mode; final selection should still rely on direct JS evaluation (the exhaustive search of §VII-A).

Three caveats remain:

1) **Correlation strength below moderate**: $|r| = 0.246$ is in the weak-to-moderate range; the best W-trial ($W = 0.288$) achieves JS $= 0.0760$, still far from the global minimum 0.01479.
2) **Chord-space mismatch**: bar-local PD is computed from a 32-step window, whose chord vocabulary is a strict subset of the full song's. The resulting PD inhabits a smaller simplicial complex.
3) **Scale sensitivity**: W_1 is sensitive to the absolute scale of birth/death coordinates, which differ between a 32-step and a 1,088-step filtration.

Relatedly, complex (simul-mixed) weighting is useful in Tonnetz conditions but degrades DFT performance; we therefore keep timeflow-only settings for DFT production runs.

VIII. CONCLUSION

We presented a unified pipeline that extracts topological structure from music via persistent homology and generates new music preserving that structure. Key findings:

1) **Distance choice is decisive but piece-dependent.** On *hibari*, DFT outperforms frequency by 38.2%, Tonnetz by 56.8%, and voice-leading by 62.4%. Across pieces, the best distance varies (DFT / Tonnetz / frequency).

2) **Model optimality follows composition character.** *hibari* favors FC; *solari* favors Transformer. Under DFT continuous input, FC is significantly better than Transformer (Welch $p = 1.66 \times 10^{-4}$).
3) **Continuous-overlap refinement is substantial.** Per-cycle τ_c (Phase 1, $\alpha = 0.5$) reaches 0.01489 ± 0.00143 ($+58.7\%$); Phase 2 ($\alpha = 0.25$) re-optimization: $\mathbf{0.01156 \pm 0.00147}$ (-22.35%, $p = 4.94 \times 10^{-11}$), and FC-cont reaches 0.00035 ± 0.00015.
4) **Bar-level generation can match full-song Phase 1.** Under Option B common window, P3_local+C gives best trial 0.0167 (below full-song 0.0213); exhaustive 32-marker search reaches 0.01479 at $m = 0$, seed 9309 (matches Phase 1 best 0.01489; $+28.0\%$ vs Phase 2 record 0.01156).
5) **Metric-objective alignment is the meta principle.** DFT is stronger for structural precision; Tonnetz is stronger for harmony-constrained variation quality.
6) **Complex/simul and Wasserstein are distance-specific.** In DFT settings, simul mixing degrades performance. Under the earlier asymmetric-window design the barcode–Wasserstein correlation collapsed ($r = -0.054$); unifying both instrument windows (Option B) restored it to $r = +0.2457$, still weaker than Tonnetz ($r = 0.503$). Both therefore require metric-conditioned use.

This work demonstrates that TDA can serve not only as an analysis tool but as a *constraint generator for musical creation*, bridging mathematical topology and compositional aesthetics.

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